SFT V11.3 V2 - Higgs Contact Ledger

Solvency Field Theory Standard Model Interface Note

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Contents

1 0. Abstract

This note upgrades the original V11.3 Higgs Contact seed into a reusable Standard Model interface for Solvency Field Theory.

The core claim is narrow:

> SFT does not replace the Standard Model Higgs mechanism. It translates Higgs-generated rest mass into rest-clock maintenance burden.

For an elementary fermion with Yukawa coupling y_f, the Standard Model gives

$$m_f = \frac{y_f v}{\sqrt{2}}.$$

SFT translates the resulting rest mass into a Compton rest-clock rate

$$\nu_{C,f} = \frac{m_f c^2}{h} = \frac{y_f v c^2}{\sqrt{2}h}.$$

Thus the Higgs vacuum expectation value sets the rest-clock rates of Higgs-coupled elementary fields. The Higgs field is not modeled as particles striking other particles. The relevant object is the nonzero Higgs background, or contact, through which rest-clock channels are maintained.

The V11.3 V2 upgrade adds four computational pieces:

1. A Higgs-contact ledger inventory for charged leptons, quarks, W, Z, and H.
2. A pair-threshold and two-clock burden table.
3. A toy electroweak switch-on table, $v \to v(T)$.
4. A temperature-derivative contact source, connecting Higgs contact to phase-transition forcing.

The companion **V11.8 SFT Baryogenesis Blender** uses this infrastructure as an application: a varying-top-Yukawa source coupled to the SFT top+$W/Z/H$ electroweak solvency forcing can leave a baryon fossil in the observed range for effective source coefficients near 10^{-8} to 10^{-7}. That result is not derived here; this note supplies the Higgs-contact machinery used by that application.

2 1. Purpose of the Note

The original V11.3 note established a conceptual bridge:

$$\text{Higgs contact} \to \text{rest mass} \to \text{Compton ticking} \to \text{ledger burden}.$$

This V2 version turns that bridge into computational infrastructure.

The goal is not to derive the Yukawa spectrum. The goal is to make the imported Standard Model masses and couplings usable in SFT accounting.

The central translation is:

$$\boxed{\text{Higgs-generated rest mass becomes a maintained rest-clock channel.}}$$

This makes the Higgs sector relevant to pair production, electroweak phase transitions, baryogenesis source windows, future Higgs VEV selection questions, and future Standard Model ledger accounting.

3 2. Guardrails

3.1 2.1 SFT does not replace the Higgs mechanism

The Standard Model relation

$$m_f = \frac{y_f v}{\sqrt{2}}$$

is imported, not replaced.

SFT adds an interpretation of the resulting rest mass as a maintained clock burden:

$$\nu_{C,f} = \frac{m_f c^2}{h}.$$

3.2 2.2 Yukawa couplings are not derived here

The Yukawa couplings y_f are inputs.

This note does not explain why

$$y_e \ll y_t.$$

It only states what follows once the Yukawa couplings and Higgs VEV are given.

3.3 2.3 Not all mass is Higgs mass

Composite hadron masses are not mostly Higgs-generated.

For protons and neutrons, most of the rest mass is QCD confinement and binding energy, not the bare Higgs-generated masses of the valence quarks.

Thus this note concerns elementary Higgs-coupled rest-clock channels, not all mass in the universe.

3.4 2.4 Photons and gluons do not have Higgs rest clocks

Photons and gluons are massless gauge bosons. They do not have rest-frame Compton clocks.

They still carry operation rate through energy:

$$\nu = \frac{E}{h},$$

but that is not a Higgs-generated rest-clock channel.

3.5 2.5 Neutrino masses are not treated as clean minimal-SM Higgs-contact entries

Neutrino mass generation is beyond the minimal Standard Model in its simplest form. Therefore neutrinos are not used here as clean Higgs-contact ledger entries.

4 3. Standard Model Higgs Contact

For charged leptons and quarks, the Standard Model Higgs mechanism gives

$$m_f = \frac{y_f v}{\sqrt{2}},$$

where m_f is the fermion mass, y_f is the Yukawa coupling, and $v \approx 246\,\text{GeV}$ is the Higgs vacuum expectation value.

SFT translates the mass into a rest-clock rate:

$$\nu_{C,f} = \frac{m_f c^2}{h}.$$

Combining the two gives

$$\boxed{\nu_{C,f} = \frac{y_f v c^2}{\sqrt{2}h}.}$$

This is the central bridge equation of V11.3.

In natural units where $c = h = 1$, the relation becomes simply

$$\nu_{C,f} = m_f = \frac{y_f v}{\sqrt{2}}.$$

In SFT language:

$$\boxed{\text{Yukawa contact sets the rate of internal rest-clock maintenance.}}$$

5 4. Bosonic Higgs-Contact Channels

The Higgs VEV also gives mass to the electroweak gauge bosons:

$$m_W = \frac{gv}{2},$$

$$m_Z = \frac{\sqrt{g^2 + g'^2}\,v}{2}.$$

The Higgs excitation mass is related to the Higgs self-coupling:

$$m_H = \sqrt{2\lambda}\,v.$$

SFT translates these into rest-clock channels as well:

$$\nu_{C,W} = \frac{m_W c^2}{h},$$

$$\nu_{C,Z} = \frac{m_Z c^2}{h},$$

$$\nu_{C,H} = \frac{m_H c^2}{h}.$$

The active electroweak/Higgs-contact sector used later is

$$\boxed{t + W/Z/H.}$$

This sector matters because it dominates the electroweak-scale Higgs-contact ledger burden.

6 5. Static Higgs-Contact Ledger Inventory

V11.3b built the static Higgs-contact ledger inventory.

For each clean Higgs-contact species, the inventory records:

$$m_i,$$

$$\nu_{C,i} = \frac{m_i c^2}{h},$$

$$2\nu_{C,i},$$

$$2m_i,$$

$$\frac{\nu_{C,i}}{\nu_{C,e}},$$

$$\frac{\nu_{C,i}}{\nu_{C,t}}.$$

The two-clock burden $2\nu_{C,i}$ is useful for pair production and annihilation accounting.

The hierarchy is enormous. The top-to-electron clock ratio is approximately

$$\boxed{\frac{\nu_t}{\nu_e} \approx 3.38 \times 10^5.}$$

In plain language:

$$\boxed{\text{one top rest clock is about 338,000 electron rest clocks.}}$$

This is the clearest numerical expression of the Higgs-contact hierarchy.

7 6. The Top + W/Z/H Active Sector

The active baryogenesis-interface sector is

$$\boxed{t + W/Z/H.}$$

V11.3b found that the rest-clock sum of this sector is approximately

$$\boxed{\nu_{t+WZH} \approx 1.13 \times 10^{26}\ \text{Hz.}}$$

The fractional split is:

Species	Fraction of $t + W/Z/H$ clock sum
t	0.368
H	0.267
Z	0.194
W	0.171

Thus the active sector is not top-only.

The top quark is the largest single channel, but the combined $W/Z/H$ contribution is about

$$0.632.$$

So the electroweak active sector is a mixed Higgs/electroweak rest-clock block:

$$\boxed{\text{top carries the largest single share; } W/Z/H \text{ carry the majority together.}}$$

8 7. Pair Production as Rest-Clock Creation

The original V11.3 note translated pair production as a conversion of spatial energy flow into internal rest-clock channels.

For a photon producing a particle-antiparticle pair,

$$\gamma \to f + \bar{f},$$

the threshold is

$$E_\gamma \geq 2m_f c^2.$$

SFT translates the new internal maintenance burden as

$$\Delta \dot{N}_{\text{rest}} = 2\nu_{C,f} = \frac{2m_f c^2}{h}.$$

Thus pair production is not only an energy threshold. It is the creation of two maintained rest-clock channels.

In compact form:

$$\boxed{\text{spatial energy stream} \to \text{two internal Higgs-contact rest clocks.}}$$

Annihilation is the reverse:

$$f + \bar{f} \to \gamma + \gamma.$$

The rest-clock channels are closed and their ledger burden is returned to radiation-like propagation channels.

This section does not claim that the Higgs field supplies the energy for pair production. The energy comes from the incoming field configuration. The Higgs-contact ledger describes the rest-clock channels that become maintainable once the pair exists.

9 8. Temperature-Dependent Higgs Contact

At high temperatures before electroweak symmetry breaking, the effective Higgs VEV is zero or suppressed.

The temperature-dependent extension is

$$v \to v(T).$$

Define

$$q(T) = \frac{v(T)}{v_0}.$$

For fixed Yukawa coupling,

$$m_i(T) = q(T)m_i(0),$$

and therefore

$$\nu_i(T) = q(T)\nu_i(0).$$

The absolute contact turn-on is

$$\frac{d\nu_i}{dx} = \nu_i(0)\frac{dq}{dx},$$

where

$$x = \ln\left(\frac{T_{\text{start}}}{T}\right)$$

is a cooling-time variable.

This is the first dynamic Higgs-contact source term:

$$\boxed{\text{Higgs-contact turn-on rate} = \text{static rest-clock rate} \times \frac{dq}{dx}.}$$

10 9. V11.3c Contact Turn-On Result

Using the toy electroweak profile adopted in the companion computations, V11.3c found that the absolute Higgs-contact turn-on

$$\frac{dq}{dx}$$

peaks near

$$\boxed{T \approx 172.8 \text{ GeV}.}$$

A gated fractional turn-on proxy peaks near

$$\boxed{T \approx 193.2 \text{ GeV}.}$$

For the active sector $t + W/Z/H$, the peak absolute turn-on rates were:

Species	$\lvert d\nu/dx \rvert$ at peak
t	7.30×10^{25} Hz
H	5.30×10^{25} Hz
Z	3.86×10^{25} Hz
W	3.40×10^{25} Hz

The total active-sector turn-on was

$$\boxed{\left| \frac{d\nu_{t+WZH}}{dx} \right|_{\text{peak}} \approx 1.99 \times 10^{26} \text{ Hz}.}$$

This is the dynamic version of the static active-sector ledger.

11 10. Varying Yukawa Contact

The most important new bridge from the baryogenesis work is that a Yukawa coupling may itself vary during a phase transition.

For the top quark, write

$$y_t(T) = y_{t,0} \left[1 + \delta_y Y(T) \right].$$

Then

$$\nu_t(T) = q(T)\frac{y_t(T)}{y_{t,0}}\nu_t(0).$$

Taking a logarithmic derivative gives

$$\boxed{\frac{d\ln\nu_t}{dx} = \frac{d\ln q}{dx} + \frac{d\ln y_t}{dx}.}$$

This is the clean V11.3-to-baryogenesis bridge.

It says that a rest-clock turn-on source may come from two places:

1. the Higgs VEV turning on,
2. the Yukawa contact changing during the transition.

V11.3c confirmed that for the toy varying-Yukawa profile,

$$\int \left|\frac{d\ln y_t}{dx}\right| dx = \ln(1+\delta_y).$$

For example:

δ_y	Integral $\int \lvert d\ln y_t\rvert$
0.2	0.182322
0.5	0.405465
1.0	0.693147

This makes the source normalization transparent.

12 11. Contact Source Versus Baryogenesis Forcing

V11.3d compared the Higgs-contact source profiles from this note with the restricted top+$W/Z/H$ forcing used in the companion baryogenesis document.

The result was:

$$\boxed{\text{pure Higgs VEV turn-on overlaps the baryogenesis forcing, but varying-top-Yukawa windows overlap better.}}$$

The pure Higgs VEV contact turn-on peaks around

$$T \approx 173 \text{ GeV},$$

while the restricted top+$W/Z/H$ baryogenesis forcing peaks lower, around

$$T \approx 137 \text{ GeV}.$$

This means pure $v(T)$ turn-on is somewhat hot/early relative to the baryogenesis forcing.

A varying-top-Yukawa source can naturally shift or broaden the source window.

The strongest tested overlap was the late moderate Yukawa source window, peaking near

$$T \approx 102 \text{ GeV},$$

with strong overlap and correlation against the V11.8 baryogenesis forcing.

The lesson is:

$$\boxed{\text{source windows slightly cooler or broader than raw VEV turn-on survive washout better.}}$$

13 12. Why This Should Remain Separate from Baryogenesis

This note should remain separate from the baryogenesis note.

The reason is structural.

V11.3 V2 is infrastructure:

$$\text{Higgs contact} \to \text{rest-clock ledger} \to \text{temperature-dependent contact source.}$$

V11.8 SFT Baryogenesis Blender is an application:

$$\frac{dY_B}{dx} = S_{\text{CP}}(T) F_{\text{SFT}}(T) - \frac{W}{H} Y_B.$$

This note supplies the particle-sector ledger and source-window machinery.

The baryogenesis note uses that machinery in a source/washout/fossil calculation.

Combining the two would dilute both. Higgs Contact is reusable across many future sectors, including pair production, electroweak phase transitions, Higgs VEV selection, Standard Model ledger inventories, baryogenesis, and future particle-sector tests.

Baryogenesis has its own guardrails and open problems.

Therefore the correct architecture is:

$$\boxed{\text{keep separate documents, strongly cross-linked.}}$$

14 13. Pointer to Companion Document: V11.8 SFT Baryogenesis Blender

This V11.3 V2 note provides the Higgs-contact machinery used by the companion document:

$$\boxed{\text{V11.8 SFT Baryogenesis Blender.}}$$

The companion document applies the contact ledger to the electroweak transition.

Its narrowed V8F claim is:

> A varying-top-Yukawa source coupled to the top+$W/Z/H$ SFT electroweak solvency forcing can produce the observed baryon-fossil scale for effective source coefficients near 10^{-8} to 10^{-7}, robustly across electroweak gate choices.

The relationship between the notes is:

$$\boxed{\text{V11.3 V2 supplies the Higgs-contact ledger; V11.8 uses it as a baryogenesis application.}}$$

15 14. What This Note Establishes

This note establishes:

1. A clean SFT translation of Higgs-generated rest mass into rest-clock burden.
2. A reusable Higgs-contact ledger inventory for Standard Model elementary species.
3. A pair-threshold and two-clock burden interpretation.
4. A temperature-dependent Higgs-contact switch-on model.
5. A dynamic contact-source equation.
6. A varying-top-Yukawa bridge equation.
7. A direct cross-link to the V11.8 baryogenesis application.

The most important equations are:

$$\nu_{C,f} = \frac{m_f c^2}{h} = \frac{y_f v c^2}{\sqrt{2}h},$$

$$\nu_i(T) = q(T)\nu_i(0),$$

$$\frac{d\nu_i}{dx} = \nu_i(0)\frac{dq}{dx},$$

$$\frac{d\ln\nu_t}{dx} = \frac{d\ln q}{dx} + \frac{d\ln y_t}{dx}.$$

16 15. What This Note Does Not Establish

This note does not establish:

1. A derivation of the Yukawa couplings.
2. A derivation of the Higgs VEV.
3. A derivation of electroweak gauge couplings.
4. A theory of neutrino masses.
5. An explanation of why there are three generations.
6. A replacement for the Standard Model Higgs mechanism.
7. A claim that all mass is Higgs-generated.
8. A completed baryogenesis theory.

The Standard Model supplies the masses and couplings. SFT supplies the ledger translation.

17 16. Open Problems

17.1 16.1 Yukawa spectrum

Why do the Yukawa couplings have their observed hierarchy?

$$y_e \ll y_t.$$

This remains open.

17.2 16.2 Higgs VEV selection

Why is

$$v \approx 246 \text{ GeV?}$$

SFT may eventually ask whether global solvency constrains the VEV, but this note does not claim that result.

17.3 16.3 Generation count

Why are there three generations?

This remains open.

17.4 16.4 Neutrino mass interface

How should SFT treat neutrino mass generation?

This requires a careful extension beyond minimal Standard Model Higgs-contact accounting.

17.5 16.5 Real finite-temperature Higgs profile

The toy $q(T)$ profile should eventually be replaced by a more realistic finite-temperature electroweak profile.

17.6 16.6 Ledger weight derivation

The bridge between rest-clock burden and full SFT ledger weights should be derived more deeply from operation counting.

18 17. Earned Statement

The following statement is earned by V11.3 V2:

> The Standard Model Higgs mechanism gives elementary particles rest masses through Higgs contact. SFT translates those rest masses into maintained Compton rest-clock channels. The resulting Higgs-contact ledger is computable, temperature-dependent, and reusable. During electroweak symmetry breaking, the turn-on of $v(T)$ and possible variation of $y_t(T)$ create calculable rest-clock source windows. These source windows form the infrastructure used by the V11.8 SFT Baryogenesis Blender.

In compact form:

$$\boxed{\text{Higgs contact is the Standard Model-to-SFT rest-clock interface.}}$$

19 18. One-Line Summary

The Higgs VEV does not merely give particles mass in SFT language; it opens and maintains elementary rest-clock channels. V11.3 V2 turns that statement into a reusable ledger inventory and a temperature-dependent source bridge for electroweak applications such as V11.8 SFT Baryogenesis Blender.

20 References and Source Notes

1. The original V11.3 seed note defined the safe Higgs-contact statement: the Higgs VEV sets rest-clock rates of elementary Higgs-coupled fields through their Yukawa couplings, while photons do not directly acquire a Higgs rest clock and composite hadron masses are not Higgs-only.
2. The numerical ledger inventory uses representative Standard Model masses and the conversion $1\,\mathrm{GeV}/h = 2.417989242 \times 10^{23}\,\mathrm{Hz}$.
3. The electroweak switch-on, varying-top-Yukawa profiles, and overlap tests are toy models intended as SFT infrastructure. They are not precision finite-temperature Standard Model calculations.
4. The companion application is **V11.8 SFT Baryogenesis Blender**, which uses this Higgs-contact ledger in a source/washout/fossil computation.